

Torsha Majumder

Kolkata, India • tmajumder.work@gmail.com • GitHub • HF • LinkedIn • Portfolio • Relocation: Singapore, Toronto, Vancouver

Professional Summary

Lead ML Architect & Research Scientist (NVIDIA Grant Recipient)

MS in Computer Science with a background in Computational Physics and 9+ years of professional experience in the USA and India. Spearheading NVIDIA-supported research to architect high-performance **SSL frameworks** for foundational-scale representation learning on massive astronomical datasets. Expert in architecting **multimodal Agentic Workflows (MAS)** and **DPO-aligned inference pipelines** (SageMaker/Docker). Specialized in bridging the gap between breakthrough scientific research and scalable industrial automation in high-stakes sectors.

Core Technical Skills

- **ML & AI (Research):** Self-Supervised Vision Transformers (DINOv2/ViT) • Representation Learning • DPO Alignment • Knowledge Distillation (BERT/MiniLM) • Anomaly Detection • NLP (spaCy, NLTK) • SciPy
- **Agentic AI & LLMs:** LangGraph • Stateful Cyclic MAS • Tool-Calling • Retrieval-Augmented Generation (RAG) • Prompt Engineering (Gemini 3.1/3.5, Llama 3, OpenAI)
- **Infrastructure & MLOps:** Inference Optimization • Parameter-Efficient Fine-tuning (PEFT) • AWS (SageMaker, S3, Boto3) • Docker/Containerization • NVIDIA Brev • MLflow • Zilliz (Milvus) • Supabase (pgvector) • Linux/Bash
- **Languages & Engineering:** Python (PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn) • C • C++ • Java • SQL (PostgreSQL, Oracle) • Flask • React.js • JavaScript
- **Data & Specialized Tools:** PySpark • Dask • LSDB • Astropy • MongoDB • Matplotlib • Git/GitHub • LaTeX

Professional & Research Experiences

Nu Energy India | Lead ML Architect | Kolkata, India

Feb. 2026 – Present

- **Architected** a proprietary **cyclic Multi-Agent System (MAS)** using **LangGraph** and **Gemini 3.1** to automate **self-correcting** industrial energy and safety auditing workflows.
- Developing a **multimodal ingestion pipeline** utilizing **Gemini Vision** and **Supabase (pgvector)** to automate entity extraction from technical manuals and instrument logs.
- Implementing a **deterministic calculation engine** using **NumPy** with a **Human-in-the-Loop (HITL)** layer, reducing audit report generation from **weeks to days** with a benchmarked **<0.5% error margin**.

The SNAD Team | Lead ML Architect (NVIDIA Grant Recipient)

Apr. 2024 – Present

- **Architected** **ASTRA**, a modular **SSL framework**; engineered **Astra-CLR**, the first **Contrastive Learning** multi-filter time-series Transformer, pre-trained on ~2.1M and **scaled to generate embeddings for ~70M ZTF** light curves for team-wide research.
- **Engineered** a **multi-view late-fusion architecture** and “**multi-crop**” paradigm within **ASTRA**; utilized a **PEFT** strategy to achieve **77% fine-tuned accuracy**; research first-authored and **submitted to A&C Journal (arXiv)**.
- Spearheading the **ASTRA ecosystem expansion** by developing a breakthrough architectural advancement targeting **top-tier ML Conferences (2027)** for a **universal scientific foundation model** for photometric data.
- Discovered **8 Superluminous Supernovae (SLSN)** candidates in the ZTF DR8 release via an **Active Learning pipeline** utilizing the “**Pineforest**” algorithm, resulting in a first-author publication in **PoS**.

Verizon | Data Scientist | Sunnyvale, CA, USA

Jun. 2021 – May 2023

- **Engineered** a scalable vendor recommendation and forecasting engine on **AWS SageMaker**, increasing prediction accuracy by **25%** through **collaborative filtering** and rigorous **A/B testing**.
- **Architected** an end-to-end automation pipeline for document classification and metadata extraction (**NER**) using **ResNet** and **OCR**, utilizing **MongoDB** and **Boto3** for **S3** integration; reducing manual operational effort by **15%**.
- **Led a 5-person cross-functional analytics team** for the **FUZE** Regulatory platform, overseeing the delivery of data-driven solutions to optimize nationwide business outcomes.

MIT | Research Affiliate | Cambridge, MA, USA

Oct. 2021 – Nov. 2023

- **Optimized the DASH spectral classification engine**, migrating the architecture from **TensorFlow** to **Keras** and implementing **batch processing** to support novel transient classes (**Kilonovae, TDEs, AGNs**).
- **Engineered unsupervised ML pipelines** for **anomaly detection** and noise removal across massive **TESS, Kepler, and PLAsTiCC** datasets to identify rare exoplanetary and transient signals.
- **Co-authored research** in astronomical deep learning published at **NeurIPS 2023**.

- Engineered a **probabilistic neural network** to optimize **MCMC solver initialization** for exoplanetary orbital modeling; second-author research submitted to the **American Astronomical Society (AAS)**.

Selected AI Portfolio

- Sentinel-Hub (Multi-Agent System):** Architected a **multimodal**, 4-agent autonomous **cyclic MAS** using **LangGraph** and **Gemini 3.1** to orchestrate cross-domain security forensics; integrated vision, transactional, and log-based **microservices** into a **stateful, self-correcting reasoning graph**.
- DeepShield (AI Safety & DPO):** Aligned a **DINOv2** backbone via a **DPO-based PEFT approach** to prioritize human-perceived risk, achieving **97.5% sensitivity** on high-fidelity fakes and a **50.6% reduction** in False Positives.
- LogSentinel (Infrastructure Security):** Engineered a **zero-day anomaly detector** using **distilled BERT** and **Zilliz (Milvus)** to perform **low-latency semantic analysis** on high-throughput HDFS logs; utilized **SSL representations** to identify semantic deviations from normal system behavior.
- NeuralAudit (Financial SSL):** Developed an **SSL-based** tabular anomaly detector using a **bottleneck autoencoder** and **Supabase (pgvector)** to identify fraudulent transactions in **high-dimensional latent space** via Euclidean distance.

Education

MS in Computer Science (Thesis) | The University of Texas at Dallas, USA | 2018 – 2020

Thesis: Ensembles of Oblique Decision Trees (ML & Algorithm Design)

Graduate Research (Physics) | The University of Lethbridge, Canada | 2024 – 2025

Transition-Edge-Sensor (TES) optimization for Double Fourier Interferometry; Instrumentation Group.

B.Tech in Information Technology | Maulana Abul Kalam Azad University of Technology, India | 2013 – 2017

Project: Institute Library Management System (DBMS)

Publications

- Majumder, T., et al. (In Prep). Large-Scale Self-Supervised Pre-training for Representational Learning in Time-Domain Astronomy.** Targeting *top-tier ML conferences (2027)*.
- Majumder, T., Malanchev, K., Ishida, E. E. O. Multi-Scale Contrastive Attention for Light-Curve Representation Learning.** Submitted to *Astronomy and Computing (A&C) Journal*, July 2026. [arXiv:2606.31627](#).
- Huang, H., ... Majumder, T., et al. Predicting the Age of Astronomical Transients from Real-Time Multivariate Time Series.** *Neural Information Processing Systems (NeurIPS 2023)*. [arXiv:2311.17143](#).
- Majumder, T., et al. Superluminous Supernova Search with PineForest.** *Proceedings of Science (PoS)*, PoS(MULTIF2025)041. Accepted Jan 2026.
- Schussler, J., Majumder, T., Penev, K. Tidal Dissipation Leading to Circularization in Kepler Binaries.** Submitted to the *American Astronomical Society (AAS) Journal*, Apr 2026.
- Majumder, T. (2020). Ensembles of Oblique Decision Trees.** *UTD Master's Thesis*. URI: hdl.handle.net/10735.1/8818.

Awards & Honors

- NVIDIA Academic Grant Recipient | NVIDIA | Oct. 2025 – Mar. 2026**
- Graduate Research Award (ULGRA) | The University of Lethbridge | 2024**
- Associate Member (AMIE) | Institute of Engineers (India), Computer Science & Engineering | 2017**
- Best Final-Year Project Award | Institute Library Management System, STCET | 2017**

Talks & Mentorships

- Selected Talks:** *MIT TESS Science Talk*, Cambridge, MA (2022); *Int'l Meeting on Eclipsing Binaries*, China/Online (2024)
- Mentorship:** Led **Data Analytics & FinTech bootcamp programs** at 2U Inc./edX for US/Australian universities (2020 – 2023).
- Instruction:** **Graduate Teaching Assistant** for Biophysics and Engineering Mechanics at The University of Lethbridge (2024 – 2025).